```
import numpy as np
import matplotlib.pyplot as plt
```

Task 1

Finish the function below, which takes a set of Fourier series coefficients, a fundamental frequency, and a vector of output times, and computes the truncated Fourier series evaluated at these times.

Recall the definition of the Fourier series:

$$f_N(t) = \Sigma_{n=-N}^{N} D_n e^{j\omega_0 nt}$$

N is based on the length of the coefficients, the equation should be

$$D_n = 2N + 1$$

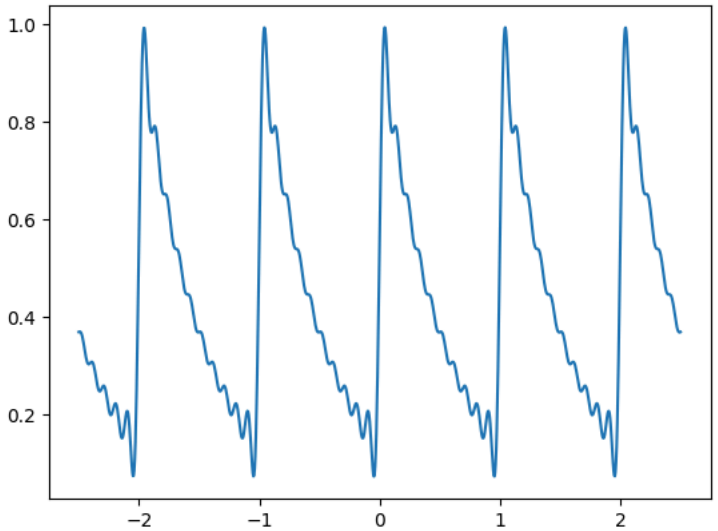
```
def my_fourier_series(coeffs, freq, times):
    """
    This function should return an output signal, when given the following:
    coeffs - Fourier series coefficients for a given signal (D_n)
    freq - The fundamental frequency of the signal (\omega_0)
    times - the time steps for which the output signal should be calculated
    """
    N = (len(coeffs) - 1) // 2
    output_fn = np.zeros_like(times, dtype=np.cdouble)
    for n in range(-N, N+1):
        D_n = coeffs[n + N]
        output_fn += D_n * np.exp(1j * freq * n * times)
    return output_fn
```

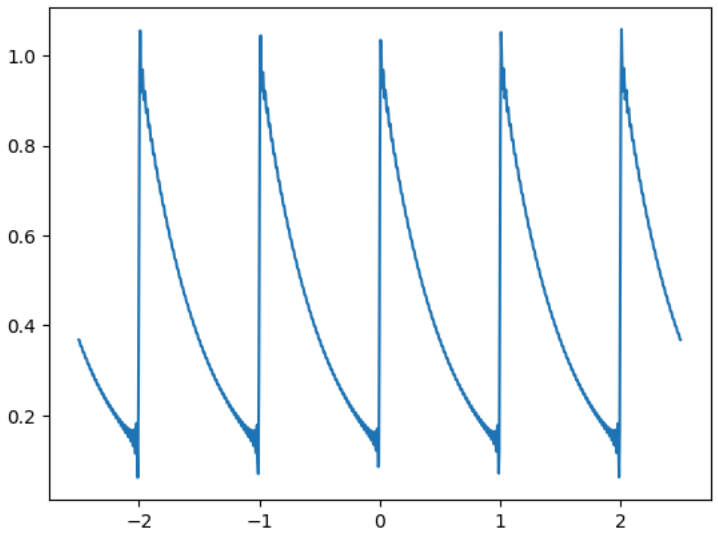
Task 2

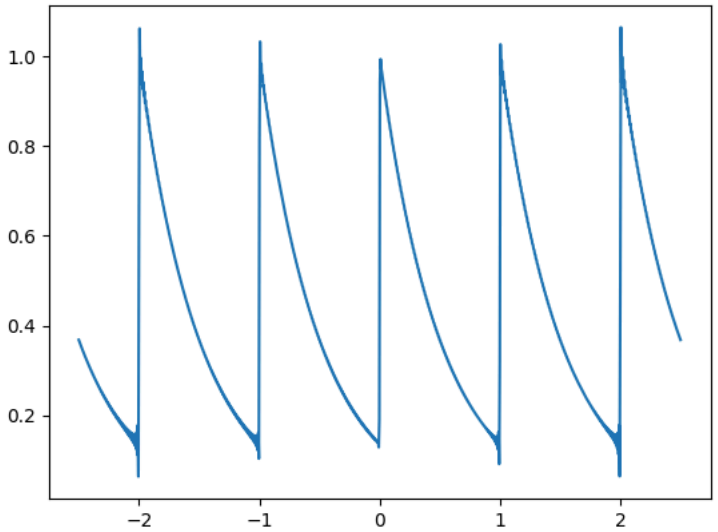
We will now verify the output of the the routine by checking the Fourier series coefficients for Problem 1-a-ii. We will do this for N=10, N=50, and N=100.

```
for N in [10, 50, 100]:
    freq = 2 * np.pi
    n = np.linspace(-N, N, 2*N+1)
    coeffs = (1-np.exp(-2))/(2+2j*n*np.pi)
    times = np.linspace(-2.5, 2.5, 1000)
    output_fn = my_fourier_series(coeffs, freq, times)
    plt.plot(times, output_fn)
    plt.show()
```

```
/usr/local/lib/python3.11/dist-packages/matplotlib/cbook.py:1709: ComplexWarning: Casting complex values to real discards the imaginary pa
  return math.isfinite(val)
/usr/local/lib/python3.11/dist-packages/matplotlib/cbook.py:1345: ComplexWarning: Casting complex values to real discards the imaginary pa
  return np.asarray(x, float)
```








```
print(0)
```

ECE 102 Homework 3

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May 15, 2025

1. (28 points) Fourier Series

(a) (18 points)

Find the Fourier series coefficients for each of the following periodic signals:

i. $f(t) = \cos(3\pi t) + \frac{1}{2}\sin(4\pi t)$

Solution:

We know that

$$f(t) = \frac{1}{2}(e^{3j\pi t} + e^{-3j\pi t}) + \frac{1}{4j}(e^{4j\pi t} - e^{-4j\pi t}) \tag{1}$$

$$= \frac{1}{2}e^{3j\pi t} + \frac{1}{2}e^{-3j\pi t} + \frac{1}{4j}e^{4j\pi t} - \frac{1}{4j}e^{-4j\pi t} \tag{2}$$

To find the period,

$$T_0 = \text{lcm}\left(\frac{2\pi}{3\pi}, \frac{2\pi}{4\pi}\right) \tag{3}$$

$$= \text{lcm}\left(\frac{2}{3}, \frac{1}{2}\right) \tag{4}$$

$$= 2 \tag{5}$$

Then $w_0 = \frac{2\pi}{T_0} = \pi$. Recall that

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\pi t} \tag{6}$$

By inspection

$$c_k = \begin{cases} \frac{1}{2} & |k| = 3 \\ \frac{j}{4} & k = -4 \\ -\frac{j}{4} & k = 4 \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

ii. $f(t)$ is a periodic signal with period $T = 1\,\text{s}$, where one period of the signal is defined as e^{-2t} for $0 < t < 1\,\text{s}$.

Solution:

By inspection, $T_0 = 1$ s, $w_0 = 2\pi$. WLOG let $0 < t < 1$. Then

$$c_k = \int_0^1 f(t)e^{-jk2\pi t}dt \tag{8}$$
$$= \int_0^1 e^{-2t}e^{-jk2\pi t}dt \tag{9}$$
$$= \int_0^1 e^{(-2-jk2\pi)t}dt \tag{10}$$
$$= \frac{e^{-2-jk2\pi} - 1}{-2 - 2jk\pi} \tag{11}$$
$$= -\frac{e^{-2} - 1}{2 + 2jk\pi} \tag{12}$$

iii. $f(t)$ is the periodic signal shown below

Solution:

By inspection, $T_0 = 3, w_0 = \frac{2\pi}{3}$. Then

$$c_k = \frac{1}{3}\int_0^3 f(t)e^{-jk\frac{2\pi}{3}t}dt \tag{13}$$
$$= \frac{1}{3}\left[2\int_0^1 e^{-jk\frac{2\pi}{3}t}dt + \int_1^2 e^{-jk\frac{2\pi}{3}t}dt\right] \tag{14}$$
$$= -\frac{2}{3jk\frac{2\pi}{3}}\left[e^{-jk\frac{2\pi}{3}t}\right]_0^1 - \frac{1}{3jk\frac{2\pi}{3}}\left[e^{-jk\frac{2\pi}{3}t}\right]_1^2 \tag{15}$$
$$= -\frac{1}{3jk\frac{2\pi}{3}}\left[2e^{-jk\frac{2\pi}{3}} - 2 + e^{-jk\frac{4\pi}{3}} - e^{-jk\frac{2\pi}{3}}\right] \tag{16}$$
$$= -\frac{1}{jk2\pi}\left[e^{-jk\frac{2\pi}{3}} + e^{-jk\frac{4\pi}{3}} - 2\right] \tag{17}$$

Using Euler's formula:

$$e^{-jk\frac{2\pi}{3}} = \cos\left(\frac{2\pi k}{3}\right) - j\sin\left(\frac{2\pi k}{3}\right) \tag{18}$$
$$= \begin{cases} 1 & k = 3n \text{ for } n \in \mathbb{Z} \\ -\frac{1}{2} - j\frac{\sqrt{3}}{2} & k = 3n+1 \text{ for } n \in \mathbb{Z} \\ -\frac{1}{2} + j\frac{\sqrt{3}}{2} & k = 3n+2 \text{ for } n \in \mathbb{Z} \end{cases} \tag{19}$$

$$e^{-jk\frac{4\pi}{3}} = \cos\left(\frac{4\pi k}{3}\right) - j\sin\left(\frac{4\pi k}{3}\right) \tag{20}$$

$$= \begin{cases} 1 & k = 3n \text{ for } n \in \mathbb{Z} \\ -\frac{1}{2} + j\frac{\sqrt{3}}{2} & k = 3n+1 \text{ for } n \in \mathbb{Z} \\ -\frac{1}{2} - j\frac{\sqrt{3}}{2} & k = 3n+2 \text{ for } n \in \mathbb{Z} \end{cases} \tag{21}$$

Thus

$$c_k = \begin{cases} 0 & k = 3n \text{ for } n \in \mathbb{Z} \\ \frac{3}{2jk\pi} & \text{otherwise} \end{cases} \tag{22}$$

(b) (10 points)

Suppose you have two periodic signals $x(t)$ and $y(t)$, of periods T_1 and T_2 respectively. Let x_k and y_k be the Fourier series coefficients of $x(t)$ and $y(t)$.

1. If $T_1 = T_2$, express the Fourier series coefficients of $z(t) = x(t) + y(t)$ in terms of x_k and y_k.
   Solution:

   Since $T_1 = T_2$, both have the same fundamental frequency as well. Let e_k, c_k and d_k be the Fourier coefficients of $z(t), x(t)$, and $y(t)$ respectively.

$$z(t) = x(t) + y(t) \tag{23}$$

$$= \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} + \sum_{k=-\infty}^{\infty} d_k e^{jk\omega_0 t} \tag{24}$$

$$= \sum_{k=-\infty}^{\infty} (c_k + d_k) e^{jk\omega_0 t} \tag{25}$$

   Thus $e_k = c_k + d_k$

2. If $T_1 = 2T_2$, express the Fourier series coefficients of $w(t) = x(t) + y(t)$ in terms of x_k and y_k.
   Solution:

   Denote the least common period by $T = \text{lcm}(T_1, T_2) = T_1$. So $w(t)$ is periodic with period T_1. $x(t)$ has period T_1 and fundamental frequency $\omega_x = \frac{2\pi}{T_1}$. $y(t)$ has period $T_2 = \frac{T_1}{2}$, so its fundamental frequency is $\omega_y = \frac{2\pi}{T_2} = 2\omega_x$

   Expand both in terms of ω_x to align them to the common period T_1. Then:

$$x(t) = \sum_{k=-\infty}^{\infty} x_k e^{jk\omega_x t} \tag{26}$$

$$y(t) = \sum_{m=-\infty}^{\infty} y_m e^{jm\omega_y t} = \sum_{m=-\infty}^{\infty} y_m e^{j(2m)\omega_x t} \tag{27}$$

Hence, we can write:

$$w(t) = x(t) + y(t) = \sum_{k=-\infty}^{\infty} x_k e^{jk\omega_x t} + \sum_{m=-\infty}^{\infty} y_m e^{j(2m)\omega_x t} \tag{28}$$

This is equivalent to a Fourier series with fundamental frequency ω_x, where the coefficient w_k is:

$$w_k = \begin{cases} x_k + y_{k/2} & \text{if } k \text{ is even} \\ x_k & \text{if } k \text{ is odd} \end{cases} \tag{29}$$

This is because y_m only contributes when $2m = k$, or when k is even.

2. (20 points) Fourier Series of Transformed Signals

Suppose that $f(t)$ is a periodic signal with period T_0, with the following Fourier series:

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} \tag{30}$$

Determine the period of each of the following signals, then express its Fourier series in terms of c_k:

(a) $g(t) = f(t) + 1$

Solution:

The period stays the same, since the constant function 1 is periodic with any period.

$$1 = \sum_{k=-\infty}^{\infty} d_k e^{jk\omega_0 t} \tag{31}$$

Thus

$$d_k = \begin{cases} 1 & k = 0 \\ 0 & k \neq 0 \end{cases} \tag{32}$$

Substituting,

$$g(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} + \sum_{k=-\infty}^{\infty} d_k e^{jk\omega_0 t} \tag{33}$$

$$= \sum_{k=-\infty}^{\infty} (c_k + d_k) e^{jk\omega_0 t} \tag{34}$$

$$\implies g_k = \begin{cases} c_k + 1 & k = 0 \\ c_k & k \neq 0 \end{cases} \tag{35}$$

(b) $g(t) = f(-t)$

Solution:

$$\begin{aligned} f(t) &= f(t + T_0) && (36) \\ \implies g(t + T_0) &= f(-(t + T_0)) && (37) \\ &= f(-t - T_0) && (38) \\ &= f(-t) && (39) \\ &= g(t) && (40) \end{aligned}$$

Thus $g(t)$ is periodic with period T_0. To find Fourier coefficients:

$$g(t) = \sum_{k=-\infty}^{\infty} c_k e^{-jk\omega_0 t} \tag{41}$$

Using change of variables,

$$g(t) = \sum_{k=-\infty}^{\infty} c_{-k} e^{jk\omega_0 t} \tag{42}$$

Thus $g_k = c_{-k}$

(c) $g(t) = f(at)$, where a is a positive real number

Solution:

$$\begin{aligned} g(t) &= g(t + T) && (43) \\ &= f(a(t + T)) && (44) \\ &= f(at + aT) && (45) \\ &\implies aT = T_0 && (46) \\ &\implies T = \frac{T_0}{a} && (47) \end{aligned}$$

Thus $g(t)$ is periodic with a period of $\frac{T_0}{a}$. $\omega_0 = \frac{2\pi}{T_0}$, and $\omega_g = \frac{2\pi}{T} = a\omega_0$. To find Fourier coefficients:

$$g(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_g t} \tag{48}$$

The coefficients c_k stay the same but the frequency spacing is scaled by a

3. (10 points) Eigenfunctions and LTI Systems

(a) (5 points) Show that $f(t) = \cos(\omega_0 t)$ is not an eigenfunction of an LTI system.

Solution:

Let $\mathcal{L}$ be a LTI system and suppose $f(t)$ is an eigenfunction. Then

$$\begin{aligned}
\mathcal{L}(f(t)) &= \mathcal{L}(\cos(\omega_0 t)) && (49)\\
&= \mathcal{L}\left(\frac{1}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t})\right) && (50)\\
&= \frac{1}{2}\mathcal{L}(e^{j\omega_0 t}) + \frac{1}{2}\mathcal{L}(e^{-j\omega_0 t}) && (51)
\end{aligned}$$

Since $\mathcal{L}$ is an LTI system, its output is the convolution of its impulse response $h(t)$ and the input signal $x(t) = e^{jw_0 t}$ (Result is the same for $x(t) = e^{-jw_0 t}$. Then

$$\begin{aligned}
\mathcal{L}(t) &= (h * x)(t) && (52)\\
&= \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau && (53)\\
&= \int_{-\infty}^{\infty} h(\tau)e^{j\omega_0(t-\tau)}d\tau && (54)\\
&= e^{j\omega_0 t}\int_{-\infty}^{\infty} h(\tau)e^{-j\omega_0\tau}d\tau && (55)
\end{aligned}$$

The right integral is a constant. Thus $e^{j\omega_0 t}$ and $e^{-j\omega_0 t}$ are eigenfunctions of LTI systems. Thus

$$\mathcal{L}(f(t)) = \frac{1}{2}e^{j\omega_0 t}\int_{-\infty}^{\infty} h(\tau)e^{-j\omega_0\tau}d\tau + \frac{1}{2}e^{-j\omega_0 t}\int_{-\infty}^{\infty} h(\tau)e^{j\omega_0\tau}d\tau \tag{56}$$

We can only factor this into

$$\frac{1}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t}) \cdot \lambda \tag{57}$$

if and only if

$$\int_{-\infty}^{\infty} h(\tau)e^{j\omega_0\tau}d\tau = \int_{-\infty}^{\infty} h(\tau)e^{-j\omega_0\tau}d\tau \tag{58}$$

This is not true in general. Thus $\cos(\omega_0 t)$ is not an eigenfunction of an LTI system.

(b) (5 points) Show that $f(t) = t$ is not an eigenfunction of an LTI system.

Solution:

$$\begin{aligned}
\mathcal{L}(t) &= (h * x)(t) && (59)\\
&= \int_{-\infty}^{\infty} h(\tau)(t-\tau)d\tau && (60)\\
&= t\int_{-\infty}^{\infty} h(\tau)d\tau - \int_{-\infty}^{\infty} \tau h(\tau)d\tau && (61)\\
&= At - B && (62)
\end{aligned}$$

This is not a multiple of t. Therefore $f(t) = t$ is not an eigenfunction of an LTI system.

4. (29 points) LTI Systems

Consider the following system:

The system takes as input $x(t)$, it first multiplies the input with e^t, then sends it through an LTI system. The output of the LTI system gets multiplied by e^{-t} to form the output $y(t)$.

(a) Show that we can write $y(t)$ as follows:

$$y(t) = \left[\left(e^t x(t)\right) * h(t)\right] e^{-t} \tag{63}$$

Solution:

By definition of an LTI system,

$$\mathcal{L}(x(t)) = h(t) * x(t) \tag{64}$$

where $h(t)$ is the impulse response of $\mathcal{L}$. Thus

$$\begin{aligned}
y(t) &= \mathcal{L}(e^t x(t))e^{-t} && (65)\\
&= [h(t) * e^t x(t)]e^{-t} && (66)\\
&= \left[(e^t x(t)) * h(t)\right] e^{-t} && (67)
\end{aligned}$$

(b) Use the definition of convolution to show that the above can be equivalently written as:

$$y(t) = \int_{-\infty}^{\infty} h'(\tau)x(t-\tau)\, d\tau \tag{68}$$

where $h'(t)$ is a function to define in terms of $h(t)$.

Solution:

From the above,

$$y(t) = \left[(e^t x(t)) * h(t)\right] e^{-t} \tag{69}$$
$$= \int_{-\infty}^{\infty} e^{\tau - t} h(t-\tau) x(\tau) d\tau \tag{70}$$

Let $\lambda = t - \tau$. Then

$$y(t) = \int_{-\infty}^{\infty} e^{-\lambda} h(\lambda) x(t-\lambda) d\lambda \tag{71}$$

Define $h'(\lambda) = e^{-\lambda} h(\lambda)$. Then

$$y(t) = \int_{-\infty}^{\infty} h'(\lambda) x(t-\lambda) d\lambda \tag{72}$$
$$= \int_{-\infty}^{\infty} h'(\tau) x(t-\tau) d\tau \tag{73}$$

(c) The above equation represents a description of the equivalent system that maps $x(t)$ to $y(t)$. Show using this that the equivalent system is LTI and determine its impulse response $h_{\text{eq}}(t)$ in terms of $h(t)$.

Solution:

Consider the input signal: $(ax_1(t) + bx_2(t))$. Then

$$\mathcal{L}(ax_1(t) + bx_2(t)) = \int_{-\infty}^{\infty} h'(\tau)[ax_1(t-\tau) + bx_2(t-\tau)] d\tau \tag{74}$$
$$= a\int_{-\infty}^{\infty} h'(\tau) x_1(t-\tau) d\tau + b\int_{-\infty}^{\infty} h'(\tau) x_2(t-\tau) d\tau \tag{75}$$
$$= a\mathcal{L}(x_1(t)) + b\mathcal{L}(x_2(t)) \tag{76}$$

Thus $\mathcal{L}$ is linear. Next consider the delayed signal $x_{t_0}(t) = x(t - t_0)$. Then

$$\mathcal{L}(x(t-t_0)) = \int_{-\infty}^{\infty} h'(\tau) x(t - t_0 - \tau) d\tau \tag{77}$$
$$= y(t - t_0) \tag{78}$$

Therefore $\mathcal{L}$ is time-invariant. Since the system is LTI, the system is the convolution of the impulse response with the signal. Thus, $y(t) = (h' * x)(t) \implies h_{\text{eq}}(t) = h'(t) = e^{-t} h(t)$. More

rigorously, using the Dirac function as the input,

$$y(t) = \int_{-\infty}^{\infty} h'(\tau)\delta(t-\tau)d\tau \tag{79}$$
$$= h'(t) \tag{80}$$
$$= e^{-t}h(t) \tag{81}$$

(d) Suppose that system S_1 is given by its step response $s(t) = r(t-1)$. Find the impulse response $h(t)$ of S_1. What can you say about the causality and stability of system S_1? What can you say about the causality and stability of the overall equivalent system?

Solution:

By definition, since the step response is the integral of the impulse response,

$$s(t) = \int_{-\infty}^{t} h(\tau)d\tau \tag{82}$$

Proof.

$$s(t) = (u * h)(t) \tag{83}$$
$$= \int_{-\infty}^{\infty} u(\tau)h(t-\tau)d\tau \tag{84}$$
$$= \int_{0}^{\infty} h(t-\tau)d\tau \tag{85}$$

Setting $\lambda = t - \tau$ and $d\tau = -d\lambda$, we get

$$s(t) = \int_{t}^{-\infty} h(\lambda)(-d\lambda) \tag{86}$$
$$= \int_{-\infty}^{t} h(\tau)d\tau \tag{87}$$

Since $s(t) = r(t-1)$, and

$$r(t-1) = \begin{cases} 0 & t < 1 \\ t-1 & t \geq 1 \end{cases} \tag{88}$$

Therefore,

$$h(t) = \frac{d}{dt}s(t) = \frac{d}{dt}r(t-1) = u(t-1) \tag{89}$$

Since the impulse response is not absolutely integrable, S_1 is not absolutely integrable. Also, since S_1 is LTI and $u(t-1)$ is not equal to 0 for all $t < 0$, it is not causal.

For the overall system

$$y(t) = e^{-t} \int_{-\infty}^{\infty} e^{\tau} x(\tau) h(t-\tau) d\tau \tag{90}$$

$$= \int_{-\infty}^{\infty} e^{\tau - t} x(\tau) u(t-\tau-1) d\tau \tag{91}$$

$u(t-(\tau+1)) = 1$ for $t \geq \tau + 1$ and 0 otherwise. Thus $\tau \leq t-1$. Thus the output becomes

$$y(t) = \int_{-\infty}^{t-1} e^{\tau - t} x(\tau) d\tau \tag{92}$$

Thus the overall system is causal. To check stability, let $|x(\tau)| \leq M$. Then

$$|y(t)| = \left| \int_{-\infty}^{t-1} e^{\tau - t} x(\tau) d\tau \right| \tag{93}$$

$$\leq \int_{-\infty}^{t-1} e^{\tau - t} \left| x(\tau) \right| d\tau \tag{94}$$

$$\leq M e^{-t} \int_{-\infty}^{t-1} e^{\tau} d\tau \tag{95}$$

$$= M e^{-t} [e^{t-1}] \tag{96}$$

$$= M e^{-1} \tag{97}$$

Thus the overall system is stable.

5. (13 points) Python Tasks